Firmware Analysis Report

Project: dlms_smart_meter_test_firmware.bin
Architecture: Unknown
File Size: 18423 bytes
SHA256: a3a52717543243be280a1cfa811a8cb59ef7f368e74ef12ded968efa665b8161

Executive Summary

This report was automatically generated by the FAWS system. It contains the results of a multi-stage static, dynamic, and symbolic firmware analysis.

Analysis Stages

Tool: STRINGS

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/strings.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Strings Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Extracted 232 strings.
Found 24 finding(s). [!] FILE_PATH: B/UH [!] FILE_PATH: DLMS/COSEM Smart Meter Firmware [!]
FILE_PATH: Protocol-Stack: IEC 62056-21 / IEC 62056-53 (DLMS/COSEM) [!] FILE_PATH:
/etc/dlms/dlms_association.conf [!] FILE_PATH: /etc/dlms/dlms_keys.conf [!] FILE_PATH:
/www/cgi-bin/meter_status.cgi [!] FILE_PATH: /www/cgi-bin/firmware_update.cgi [!] FILE_PATH:
/opt/meter/scripts/ota_update.sh [!] IPv4_ADDRESS: http://192.168.1.1:4059/dlms/association [!]
URL: http://192.168.1.1:4059/dlms/association Scan complete.
```

Tool: BINWALK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/binwalk.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Python Binwalk Engine on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 3 finding(s). [!]
Offset 0x0: ELF 32-bit LSB executable [!] Offset 0x686: gzip compressed data [!] Offset 0x6c5:
Squashfs filesystem Scan complete.
```

Tool: CUTTER

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/cutter.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Cutter (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 1 finding(s). [!] Architecture: ELF executable detected Scan complete.
```

Tool: GHIDRA

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/ghidra.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Ghidra (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 2 finding(s). [!] Compiler: GCC compiler artifacts detected [!] Functions: x86_64 function prologues identified Scan complete.
```

Tool: TRUFFLEHOG

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/trufflehog.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Secret Scan on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 3 finding(s). [!] GENERIC_SECRET at line 25: password=admin123 [!] AWS_KEY at line 28: aws_access_key_id = AKIAIOSFODNN7EXAMPLE [!] GENERIC_SECRET at line 29: token=fakeTestToken_1234567890abcdef Scan complete.
```

Tool: ENTROPY

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/entropy.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Entropy Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Overall Shannon Entropy: 7.9705 Found 1 finding(s). [!] High entropy (7.97) indicates packed or encrypted firmware section Scan complete.
```

Tool: WIRESHARK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/wireshark.py --target

"C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Wireshark (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 2 finding(s). [!] Network: HTTP traffic signatures found [!] Network: SSH traffic signatures found Scan complete.
```

Tool: AFL++

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/afl.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting AFL++ (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 2 finding(s). [!] Fuzzing: Potential format string attack surface detected [!] Fuzzing: Usage of unsafe string functions detected (strcpy/gets) Scan complete.
```

Tool: ANGR

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/angr.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting angr (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin... Found 1 finding(s). [!] Symbolic: High number of branch instructions (530) indicates complex control flow suitable for symbolic execution Scan complete.
```

Tool: SCORECARD

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/scorecard.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0

```
Starting Risk Scorecard Generation on project 23... Found 1 finding(s). [!] RiskScore: Calculated Aggregate Risk Score: 0/100 based on 25 findings Scan complete.
```

Tool: PDF_REPORT

Status: RUNNING | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/pdf_report.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\23_dlms_smart_meter_test_firmware.bin" --project "23" --run-id 0